

Homework 1

Macroeconomic Theory I

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1 Labor Market Data

1.1 Two-state hazards from duration shocks

1.2 The Flow Steady State

2 Theory

2.1 McCall with Separation

2.1.1

We begin by writing the Bellman equations. Unemployed workers cannot separate from jobs, so their value function is exactly the same as in the base model:

$$U = b + \beta \max_w \{E(w), U\} = b + \beta \left[\int_{w^*}^{\bar{w}} E(w) dF(w) + \int_w^{w^*} U dF(w) \right]$$

The change in this model lies in the value function for the employed worker $E(w)$, which is no longer the value of receiving w forever. It is now given by

$$E(w) = w + \beta[(1 - \sigma)E(w) + \sigma U]$$

which should remain constant for a given wage w since there is no renegotiation or on-the-job search. Since it is constant, we can rearrange for this value:

$$(1 - \beta(1 - \sigma))E(w) = w + \beta\sigma U \Rightarrow E(w) = \frac{w + \beta\sigma U}{1 - \beta(1 - \sigma)}$$

The reservation wage is the wage at which $E(w^*) = U$, so we can also write

$$U = \frac{w^* + \beta\sigma U}{1 - \beta(1 - \sigma)} \Rightarrow (1 - \beta(1 - \sigma))U = w^* + \beta\sigma U$$

We can then see that

$$U - \beta U + \beta\sigma U - \beta\sigma U = w^* = (1 - \beta)U \Rightarrow U = \frac{w^*}{1 - \beta}$$

which is exactly the same as in the original model! This is because the value of U does not depend directly on the separation probability; unemployed workers don't separate from jobs. The reservation wage encodes all the effects of the separation hazard.

Now that we have solved for E and U , we can solve for the reservation wage itself. Going back to the unemployed worker's Bellman equation, we now have:

$$\frac{w^*}{1 - \beta} = b + \beta \int_{w^*}^{\bar{w}} \frac{w + \beta\sigma \frac{w^*}{1 - \beta}}{1 - \beta(1 - \sigma)} dF(w) + \beta \int_w^{w^*} \frac{w^*}{1 - \beta} dF(w)$$

From this point we have no choice but to grind through algebra.

2.1.2

2.1.3

2.2 The wage equation, in and out of steady state

3 Computation

3.1 Calibrate to your hazards

3.2 Comparative statics

3.3 The Beveridge loop